

```
1 interface arb_if #(parameter N = 4) (input bit clk);
2
3     logic [N-1:0] req;
4     logic [N-1:0] gnt;
5     bit rst;
6
7
8     modport TEST (
9         output req,rst,
10        input  gnt,clk
11    );
12
13
14    modport DUT (
15        input  req,clk,rst,
16        output gnt
17    );
18
19
20    modport MON(
21        input req,gnt
22    );
23
24 endinterface
```

```

1 module arbiter #(parameter N = 4)(
2     arb_if.DUT arb_bus
3 );
4
5     logic [$clog2(N)-1:0] ptr;
6     int index;
7
8     always_ff @(posedge arb_bus.clk or posedge arb_bus.rst) begin
9         if (arb_bus.rst) begin
10             ptr <= 0;
11             arb_bus.gnt <= 0;
12         end else begin
13             arb_bus.gnt <= 0;
14             for (int i = 0; i < N; i++) begin
15                 index = (ptr + i) % N;
16                 if (arb_bus.req[index]) begin
17                     arb_bus.gnt[index] <= 1;
18                     ptr <= index + 1;
19                     break;
20                 end
21             end
22         end
23     end
24
25 endmodule

```

```

1 module test #(parameter N = 4)(
2     arb_if.TEST arb_bus
3 );
4
5     logic [N-1:0] req;
6     logic [N-1:0] gnt;
7     bit rst;
8
9     assign clk = arb_bus.clk;
10    assign gnt = arb_bus.gnt;
11    assign arb_bus.rst = rst;
12    assign arb_bus.req = req;
13
14    initial begin
15        rst = 1;
16        #10 rst = 0;
17
18        req = 4'b0001;
19        repeat (5) @(negedge clk);
20
21        req = 4'b1010;
22        repeat (5) @(negedge clk);
23
24        req = 4'b0000;
25        repeat (3) @(negedge clk);
26
27        $stop;
28    end
29

```

```
1 module monitor(  
2   arb_if.MON  arb_bus  
3 );  
4  
5  
6 initial begin  
7   $monitor("[%0t] REQ = %b | GNT = %b", $time, arb_bus.req, arb_bus.gnt);  
8 end  
9  
10  
11 endmodule █
```

```
1 `include "interface.sv"
2 `include "test.sv"
3 `include "monitor.sv"
4
5 module top;
6
7     parameter N = 4;
8     bit clk;
9
10    initial begin
11        clk = 0;
12        forever begin
13            #1ns clk = ~clk;
14        end
15    end
16
17    arb_if #(N) arb_bus (clk);
18    arbiter #(N) arb_dut (arb_bus.DUT);
19    test      #(N) tb      (arb_bus.TEST);
20    monitor          mon      (arb_bus.MON);
21
22 endmodule
_
```